

Log Hacker Information Using Azure Sentinel

Group 4

Elvis Tang
Sriman Komaragiri
Raaghav Wadhawan
Tyson Duran

Table of Contents

ABSTRACT:.....3

INTRODUCTION:.....4

SETTING UP AZURE:.....5

CREATING AND CONFIGURING THE HONEYPOT VIRTUAL MACHINE:..... 5

SCRIPTING THE SECURITY LOG EXPORTER..... 8

CREATING AND CONFIGURING LOG ANALYTICS WORKSPACE:.....9

CREATING A CUSTOM LOG:..... 10

CONFIGURING AZURE SENTINEL (SIEM):.....10

MACHINE LEARNING WORKSPACE:.....15

EXPERIMENTAL METHODOLOGY.....17

EXPERIMENTAL RESULTS..... 18

SUMMARY OF CHALLENGES AND HOW THEY WERE SOLVED..... 19

CONCLUSION..... 20

REFERENCES..... 21

ABSTRACT:

For our project, we will be exploring the concept of Security Information and Event Management, or SIEM, which is a security solution that helps organizations recognize potential security threats and vulnerabilities before they have a chance to disrupt business operations. For our project, we will conduct an experiment using a SIEM solution native to Microsoft Azure to log attacker information and present a map portraying the geolocation of these attackers. Though our project will operate on a smaller scale than large companies, it still presents us with an understanding of what is going on with any system vulnerabilities to then further diagnose and create potential solutions to prevent them from happening. Since this project is operating within a smaller time frame, we will need a much more exposed virtual machine to make it enticing for hackers to attack it. This virtual machine will contain a Honeypot, which acts as a decoy to lure hackers into detection and prevent them from entering the system. As a result, our project acts as a small-scale analyzer for system vulnerabilities and displays information of attacks in real-time to help us diagnose security threats and vulnerabilities before they have a chance to disrupt system operations.

INTRODUCTION:

The objective of this project is to enhance the security posture of an organization by leveraging the capabilities of Security Information and Event Management (SIEM) solutions, specifically Microsoft's Azure Sentinel. SIEM is a comprehensive approach to security operations that allows organizations to collect, analyze, and act on security data from a variety of sources, including on-premises, cloud, and hybrid environments. By implementing Azure Sentinel, the organization can proactively detect and respond to cyber threats in real-time, minimizing the potential impact of security incidents. Furthermore, by using a honeypot to bait attackers, the organization can gain valuable insights into the type and frequency of attacks targeting their assets, allowing for custom alerting and response strategies.

Azure Sentinel MAP (Microsoft Azure Sentinel Monitoring and Alerting Platform) takes this a step further by providing a live visualization of cyber-attacks as they occur. The MAP dashboard displays a map of the world with markers indicating the location of attacks in real-time. This information can be used to quickly identify and respond to potential threats and to take proactive measures to prevent future attacks.

One unique aspect of Azure Sentinel MAP is that it can be used in conjunction with a honeypot to bait hackers into attempting to log into a virtual machine. To make the honeypot more enticing to hackers, the virtual machine's firewall is disabled. After the hackers attempt to log in and fail, their location and IP are logged onto the Log Analytics Workspace using Windows PowerShell. This is then sent to a third-party API to receive their latitude, longitude, state/province, and country. We then set up Azure Sentinel to map their geolocation. This is visualized on the MAP dashboard, which provides a live view of the location and number of attempted attacks.

Using Azure Sentinel MAP with a honeypot allows security operations teams to gain valuable insight into the type and frequency of cyber-attacks targeting their organization. This information can be used to create custom alerts and responses to protect against future attacks. The visualization provided by Azure Sentinel MAP allows security teams to quickly identify and respond to threats in real-time, thereby reducing the potential damage caused by cyber-attacks.

Overall, the objective of this project is to increase the organization's ability to protect their assets and data against cyber threats, thereby minimizing the risk of financial loss, reputational damage, and legal liabilities. Azure Sentinel is a cloud-native security information and event management (SIEM) solution that provides a comprehensive approach to security operations. With Azure Sentinel, organizations can easily collect, analyze, and act on security data from a variety of sources, including on-premises, cloud, and hybrid environments. One of the most exciting features of Azure Sentinel is the integration with Microsoft Threat Intelligence, which provides real-time insights into the latest threats and vulnerabilities.

SETTING UP AZURE:

Microsoft Azure is a cloud computing platform service provided by Microsoft. It offers a full range of cloud-based tools and services that let businesses create, launch, and manage apps and services through data centers run by Microsoft.

Infrastructure as a Service (IaaS), Platform as a Service (PaaS), and Software as a Service (SaaS) options are just a few of the many services that Azure provides. It offers a wide range of services, including virtual machines, storage options, databases, networking capability, analytics tools, artificial intelligence (AI) services, Internet of Things (IoT) services, and much more.

A resource group in Microsoft Azure is a logical container that holds related resources for an application or solution. It is a collection of resources that share the same lifecycle, permissions, and policies. In our case, the resource group will hold resources for our virtual machine, our log analytics workspace, and our Azure Sentinel or SIEM. A resource group will apply consistent policies, permissions, and tags to our virtual machine, log analytics workspace, and SIEM. We will have a default of \$200 credit for each Azure subscription, which is more than enough for our project. The name of our resource group for this project will be honeypotlab.

CREATING AND CONFIGURING THE HONEYPOT VIRTUAL MACHINE:

A honeypot virtual machine in Azure serves as a valuable tool for cybersecurity professionals, researchers, and educators to gain real-world insights into the tactics, techniques, and procedures (TTPs) employed by malicious actors. The concept of a honeypot involves setting up a virtual machine or network with intentionally weak security measures, making it an enticing target for potential attackers.[2]

The primary purpose of a honeypot is to divert and capture the attention of hackers, enabling the collection of valuable data regarding their methods, motivations, and the vulnerabilities they exploit. By creating an environment that mimics legitimate systems or services, the honeypot virtual machine acts as a decoy, luring attackers away from critical infrastructure and sensitive data.

The honeypot can be designed to emulate various systems, such as web servers, database servers, or IoT devices, depending on the specific research or educational goals. It can simulate vulnerabilities, misconfigurations, or outdated software versions that are commonly targeted by attackers. This provides a controlled environment for the study and analysis of their behavior, enabling the identification of emerging threats and the development of countermeasures.[3]

Researchers and educators can gather valuable information from the honeypot, including the methods used to gain unauthorized access, the tools and malware employed, the techniques for lateral movement within the network, and the activities performed once inside the compromised system. These insights can aid in improving threat detection and response strategies, enhancing incident response capabilities, and bolstering overall cybersecurity defenses.

It is important to note that the honeypot virtual machine should be isolated from the production environment to prevent any potential impact on critical systems or data.[2] Additionally, regular monitoring and analysis of the honeypot are essential to ensure that the captured data is utilized effectively and to keep the honeypot updated with the latest vulnerabilities and attack vectors.

By deploying honeypot virtual machines in Azure, security professionals can actively study and understand the evolving nature of cybersecurity threats, providing invaluable knowledge for proactive defense measures. Furthermore, the insights gained from honeypot deployments can contribute to the development of new security technologies, threat intelligence, and training programs, strengthening the global cybersecurity community's collective resilience against malicious activities.[3]

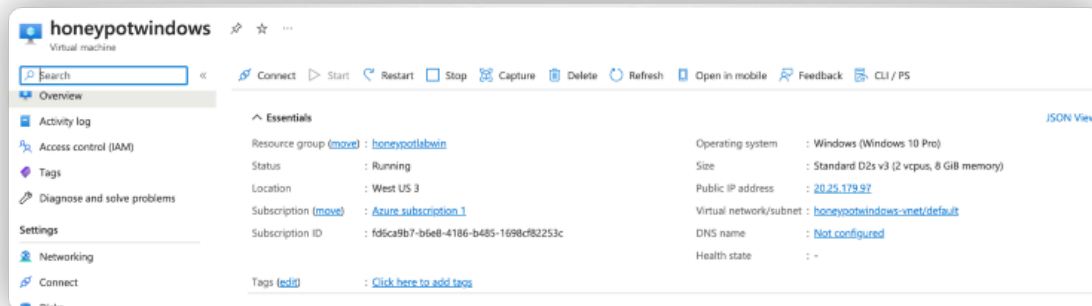


Figure 1: Deploying honeypot in Azure

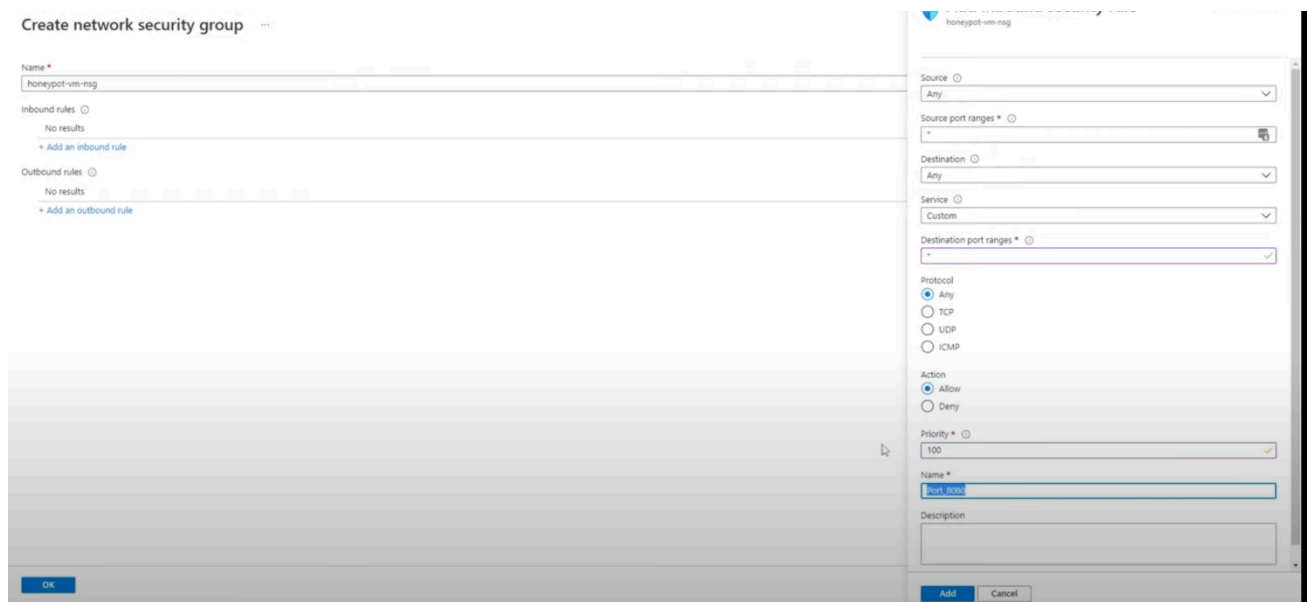


Figure 2: Disabling the Firewall in the Virtual Machine

To make our honeypot more enticing for attackers, we want our virtual machine to be as vulnerable as possible. The best way to do this is disabling the firewall in our Windows machine. We will remove the protection provided by the built-in Windows Firewall, which is a security feature that helps protect computers from unauthorized access and malicious network traffic.

When disabled, there are no active restrictions or filtering rules in place for incoming and outgoing network traffic. Our virtual machine will be exposed to potential network threats due to incoming and outgoing traffic no longer being filtered or restricted.

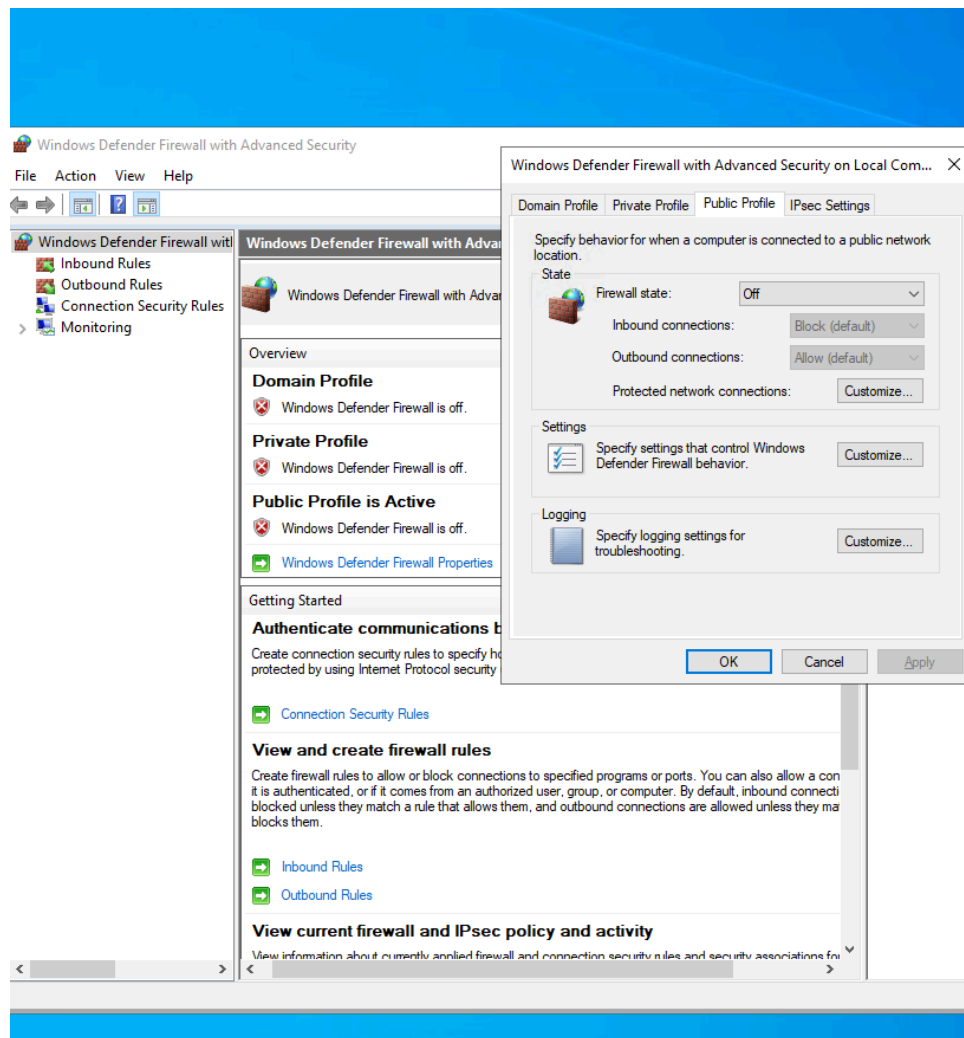


Figure 3: Disabling the Firewall in the Virtual Machine

The steps to remove our Windows Firewall are as follows: opening Windows Defender Firewall, clicking on Windows Defender Firewall Properties, and turning Firewall State to Off for our Domain Profile Private Profile and Public Profile. In general, it is not recommended to disable your firewall to reduce potential risks and implications as

it can expose your computer to security vulnerabilities.

SCRIPTING THE SECURITY LOG EXPORTER

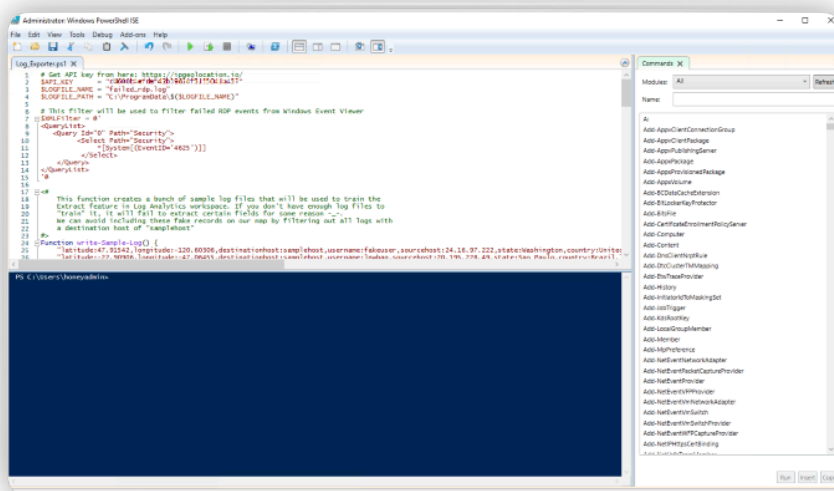


Figure 4: Powershell Script

PowerShell is a scripting and automation framework that provides a command line shell and scripting language specifically designed for system administration tasks, configuration management, and task automation. Our purpose for PowerShell in our project is to read our Event Viewer Logs in our virtual machine, specifically the failed log in attempts which are Event ID 4625. With our failed login attempts identified, we will use the IP address from the failed attempt and send that IP to an IP geolocation API called ipgeolocation.io. We get 1000 calls a day to the API for free, but we found that this was not enough with the amount of failed login attempts we got in our project. One of our project members bought the API key for 15 dollars to allow 150,000 API calls per month, which is plenty for our project.

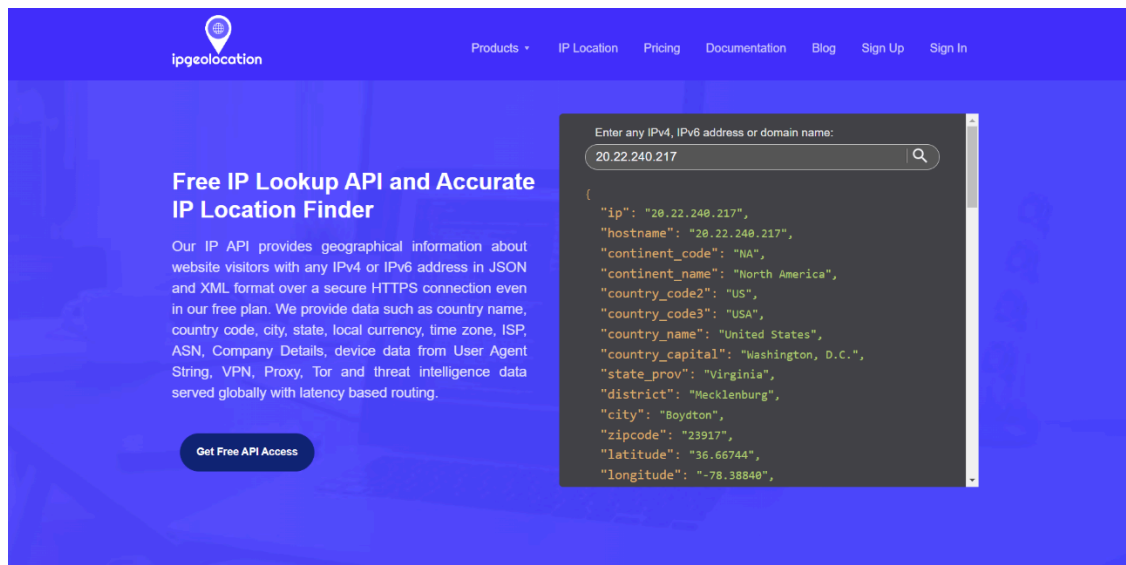


Figure 5: Geolocation API

When running the PowerShell script, it will output a log to a place on our desktop in our virtual machine. This will contain the raw data from the failed login attempts while the script was running. This log will contain the IP address, the username the attacker used, the latitude, longitude, country, and state. This log will be mapped to our Log Analytics Workspace, which will collect the log data, and use this log data for our visual map.

CREATING AND CONFIGURING LOG ANALYTICS WORKSPACE:

In our walkthrough, we create and configure a Log Analytics workspace in Microsoft Azure. The process begins with signing into the Azure portal and navigating to the "Log Analytics workspaces" tab. From there, we click "Add" and are prompted to choose a subscription, resource group, and workspace name. Next, we select the desired region and pricing tier, with options ranging from free to enterprise-level plans. Additionally, we demonstrate how to configure data sources and workspace data retention settings to ensure that the workspace is collecting and storing the necessary logs and data. Once the settings are confirmed, we click "Create" to deploy the Log Analytics workspace.

Throughout our walkthrough, we highlight key features and tools available in the Log Analytics workspace, such as the ability to add data sources from a variety of services and platforms. The user can also create custom queries using the Azure portal's query editor to retrieve specific data from the logs collected in the workspace. Additionally, we briefly discuss other Log Analytics features, including alerts and dashboards, which allow users to set up automated notifications based on specific criteria and to visualize data in a customizable dashboard format. Overall, our walkthrough provides a comprehensive overview of how to set up and use a Log Analytics workspace in Azure, highlighting the importance of data collection and analysis in monitoring and optimizing cloud-based systems.

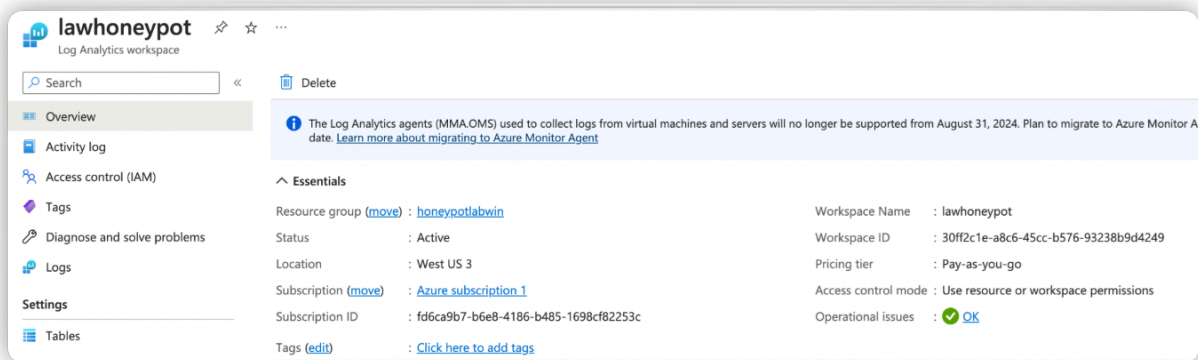


Figure 6: Log Analytics Workspace

CREATING A CUSTOM LOG:

We need a way to train Log Analytics Workspace so it can filter out specific attributes such as latitude, longitude, country, state, IP address, etc. The log we will use to train the workspace will be called `FAILED_RDP_WITH_GEO_CL`. This log is mapped to the output log from our script in our virtual machine, that includes the geolocation from our attackers.

From here, we can extract the fields from our custom log in Log Analytics Workspace such as our latitude, longitude, destination, country, and state. These fields will be used when we map out our data onto our visual map with Azure Sentinel or SIEM.

To extract the data from our log results, we will run the log '`FAILED_RDP_WITH_GEO_CL`' and extract the fields from one of the results. From here we will create a Field Title for the name of the field, select the data type for that field, and extract and save the field. This will look at the other results from the custom log and apply the fields we created to the raw data. Some of the fields we created were longitude, latitude, country, and state fields. These fields will help train Log Analytics Workspace to create the map in Sentinel.

CONFIGURING AZURE SENTINEL (SIEM):

The purpose of Azure Sentinel Map is to provide security operations teams with a visual representation of attack locations based on IP addresses, allowing them to gain valuable insights into the geographical origins of cyber-attacks. By mapping out the different places from where attacks are originating, security analysts can better understand the distribution and patterns of malicious activities, enabling them to make informed decisions and take proactive measures to protect their organization's assets and data. The primary benefit of Azure Sentinel Map is its ability to present a real-time, interactive visual representation of attack locations. This empowers security teams to quickly identify trends, hotspots, and emerging threat vectors. By visualizing attack locations on a map, analysts can easily identify geographical areas that are experiencing a higher concentration of attacks, helping them prioritize response efforts and allocate resources accordingly.[5]

Azure Sentinel Map allows for deeper analysis and correlation of attack data with other security information. By overlaying attack locations on top of other security metrics, such as log data, threat intelligence feeds, or historical attack patterns, security teams can gain deeper insights into the context and nature of the attacks. This comprehensive understanding enables them to develop effective strategies to detect, mitigate, and prevent future attacks. Azure Sentinel Map facilitates enhanced situational awareness. Security analysts can monitor the map in real-time and receive alerts or notifications when new attack locations are detected. This timely information empowers them to respond swiftly to emerging threats, investigate suspicious activities, and initiate appropriate incident response procedures. By visually tracking attack locations, security teams can improve their incident response time, minimizing the impact of attacks and reducing the potential for financial losses or reputational damage.[3] Azure Sentinel Map serves as a powerful tool for security operations teams by providing a visually intuitive representation of attack locations based on IP addresses. It offers a real-time view of the geographical distribution of attacks, enabling security analysts to identify trends, prioritize response efforts, and make informed decisions to protect their organization's assets and data. By leveraging this visual insight, security teams can enhance situational awareness, improve incident response capabilities, and strengthen their overall cybersecurity defenses.

We created an Azure Sentinel instance in your Azure portal, which will serve as the central hub for collecting and analyzing security logs. Once Azure Sentinel is set up, navigate to the "Data connectors" section and enable the "Azure Log Analytics" connector. Link it to your Log Analytics Workspace that contains the security logs. To enable IP geolocation in Azure Sentinel Map, configure IP geolocation lookup in your Log Analytics Workspace. Go to the "Advanced Settings" of the workspace, select "Data," and then "Custom Logs." Add a new custom log with the log type "CustomLogs_CL" and use the query to retrieve the necessary IP geolocation data from the AzureDiagnostics table. Once configured, Azure Sentinel Map will be able to read the logs from the Log Analytics Workspace and map out the different attack locations based on their IP addresses, providing visual insights into the geographic origins of the attacks.

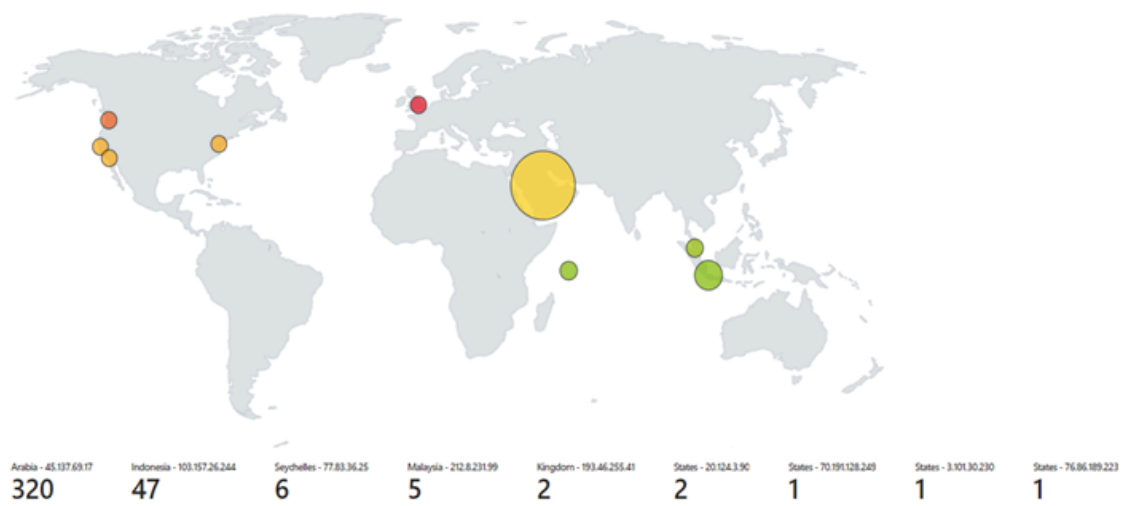


Figure 7: Trail 1 Day 1

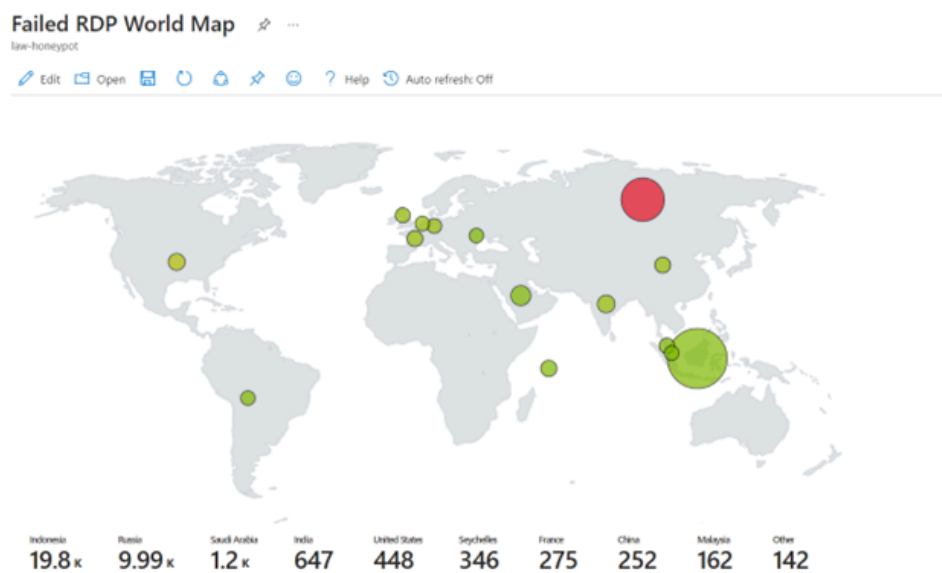


Figure 8: Trail 1 Day 2

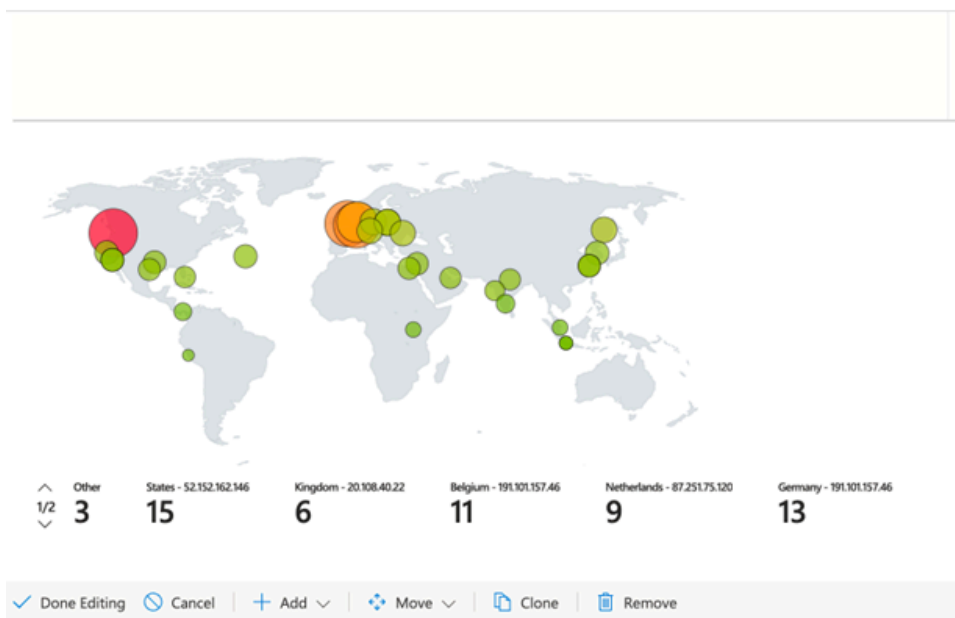


Figure 9: Trail 2 Day 1



Figure 10: Trail 2 Day 2

Failed RDP World Map

law-honeypot1

Edit Open Auto refresh: 5 minutes



Vietnam - 116.96.44.244
718

United States - 76.170.202....
1

United States - 71.6.199.23
1

Figure 11: Trail 3 Day 1

Failed RDP World Map

law-honeypot1

Edit Open Auto refresh: Off



1/2 Vietnam - 116.96.44.244
2.59 k

South Korea - 221.150.432...
1.28 k

Taiwan - 120.127.252.162
760

South Korea - 115.92.155.19
519

South Kore
460

Figure 12: Trail 3 Day 2

MACHINE LEARNING WORKSPACE:

Creating a machine learning model in a Security Information and Event Management (SIEM) system using training data from Azure Log Analytics Workspace can help identify and prevent cyber-attacks. With over 10,000 real-time cyber-attack data from the log analytics workspace, we can develop a machine learning model that detects whether an attacker is a bot or an actual person. The purpose of the machine learning model is to determine this based on the frequency of the IP address and the username used to brute force into the honeypot virtual machine in Azure.[4] To begin, we can use Python and Pandas library to read the CSV file containing the training data into a 2-dimensional list. After that, we can create a Pandas data frame, index the columns we need for our analysis, and remove the irrelevant ones. Then, we can group the data by source host and username to determine the frequency of each.

We can then create a list of data frames for each username, and group them by source host to obtain the frequency of attacks. This data can be plotted using Matplotlib library to help visualize the results. By analyzing the data, we can train the machine learning model to detect patterns and predict whether an attacker is a bot[4] or a person based on the frequency of IP addresses and usernames used to attempt brute force attacks.

In addition to the programming aspects of using Python and Pandas to analyze the data from the Azure Log Analytics Workspace, we used Scikit-Learn, a popular Python library for machine learning, to build a Decision Tree Classifier model to classify incoming traffic as either bot or human.

The Decision Tree Classifier model is a supervised learning algorithm that uses a tree-like structure to model decisions and their possible consequences.[6] In our case, we can train the model on the data from the Azure Log Analytics Workspace, with the frequency of IP addresses and usernames as the features and the labels as either bot or human. Once trained, the model can classify incoming traffic based on the features and predict whether it's a bot or human.

However, we encountered the problem of having too many bots and not a lot of human hackers in our training data, leading to overfitting of the model. Overfitting occurs when a model is too complex, leading to it performing well on the training data but poorly on new, unseen data. In our case, the model was too sensitive to the training data, leading to it misclassifying new data.

To address this, we could have tried several approaches. First, we could have balanced the training data by oversampling the human hacker data or under sampling the bot data to ensure that the model has enough examples of both classes. We could have also used cross-validation techniques to test the model's performance on new, unseen data to ensure that it's not overfitting to the training data.

Another approach would have been to use ensemble methods such as Random Forest[4], which combines multiple decision tree models to improve prediction accuracy and reduce overfitting. By training multiple decision tree models on different subsets of the data, it would have been possible reduce the variance in the model and improve its performance on new data.

In the end, theoretically, it would have still been possible to create this model using the decision tree classifier if there were more samples that had actual human hackers. All in all, using machine learning in a SIEM system can significantly enhance the ability to detect and prevent cyber-attacks. By analyzing the data from the Azure Log Analytics Workspace, we can develop a robust machine learning model that can identify and prevent malicious activities in real-time. The insights and predictions provided by the model can help organizations improve their cybersecurity defenses, protecting their critical assets and infrastructure from potential threats.

sourcehost_CF				
141.98.83.121	1687			
185.7.214.232	1624			
103.123.66.130	1304			
45.95.202.179	758			
141.98.83.88	733			
185.217.199.66	670			
185.104.251.129	652			
185.188.183.59	575			
185.104.251.153	574			
27.71.85.174	323			
Name: sourcehost_CF, dtype: int64				

	precision	recall	f1-score	support
0	1.00	1.00	1.00	13043
1	0.99	0.99	0.99	4343
accuracy			0.99	17386
macro avg	0.99	0.99	0.99	17386
weighted avg	0.99	0.99	0.99	17386
decision tree				
adaboost				

Figure 13&14: Prediction Model GUI

EXPERIMENTAL METHODOLOGY

Furthermore, we made the honeypot more vulnerable by disabling the firewall in the virtual machine, which is a security feature that helps protect computers from unauthorized access and malicious network traffic. It further recommends the use of PowerShell, a scripting and automation framework, to read the Event Viewer Logs in the virtual machine, specifically the failed log-in attempts. We used IP geolocation API to get the location of the IP address from the failed attempt.

Moreover, a step-by-step guide on creating and configuring the honeypot virtual machine in Azure is provided, including setting up a logical container or resource group that holds related resources for an application or solution, such as virtual machines, log analytics workspace, and SIEM. The resource group applies consistent policies, permissions, and tags to the virtual machine, log analytics workspace, and SIEM. As stated before, each Azure subscription has a default \$200 credit, which is more than enough for the project.

Finally, the paper emphasizes the importance of regular monitoring and analysis of the honeypot to ensure that the captured data is utilized effectively and to keep the honeypot updated with the latest vulnerabilities and attack vectors. It concludes that deploying honeypot virtual machines in Azure provides invaluable knowledge for proactive defense measures, and the insights gained can contribute to the development of new security technologies, threat intelligence, and training programs, strengthening the global cybersecurity community's collective resilience against malicious activities.

EXPERIMENTAL RESULTS

From our results above, we collected many tests as we showed in the form of screenshots, event viewer logs, logs from Azure, the Azure Sentinel Map, and the machine learning model training results. We also include explanations for each picture of what it is about.

For creating and configuring the honeypot virtual machine, we include the screenshots of the virtual machine, screenshots for checking our firewall rules in the virtual machine, and the inbound rules set in Azure.

For creating and configuring Log Analytics Workspace, we include the images of the queries in Azure, and the PowerShell code to retrieve hacker data.

For the Azure Sentinel results, we include images of the maps of all three trials. Each trial shows the map on Day 1 and the same map on Day 2, illustrating the dramatic influx of hackers, both bots and humans.

For the machine learning workspace, we include the images of the model's training results and the preprocessing data of the number of hackers in certain IP addresses. This shows us the overfitting of the model, but we further explain as to why that is an issue and how it can be changed with further analysis and time on this project.

SUMMARY OF CHALLENGES AND HOW THEY WERE SOLVED

Creating the sentinel map came with many challenges. The most challenging thing was to get familiar with and experience Microsoft Azure and PowerShell. No one in the group had prior experience with these technologies. To combat this problem, the team spent the first half of the semester researching and experimenting with Microsoft Azure and PowerShell scripts/coding. While working on our Log Analytics Workspace, we ran into a drawback with Microsoft Azure's built-in machine learning tool. We used this tool to extract specific data from potential hackers or security threats to our system. This data included attributes such as longitude, latitude, country, IP address, etc. The problem we ran into is sometimes the tool would not extract the proper data for a certain attribute. When that problem came about, we had to manually go back and reset that attribute so we could get precise data. Another aspect of the project which seemed to be inaccurate was the geolocation tracking for the map. We noticed that the utilization of the longitude and latitude attributes for the location on the map was inaccurate, so we changed the mapping settings to be based on the country attribute. Lastly, something we could have improved on in this project was in our machine learning model. The machine learning model was overfitting. This means the model gets too used to individual computer data, which unfortunately does not allow the data to be used universally, or on any machine.

CONCLUSION

In conclusion, the implementation of Azure Sentinel in the organization's security infrastructure significantly enhances their security posture. By leveraging the capabilities of SIEM solutions, specifically Microsoft's Azure Sentinel, the organization can proactively detect and respond to cyber threats in real-time, minimizing the potential impact of security incidents. Furthermore, by incorporating a honeypot to bait attackers, the organization gains valuable insights into the type and frequency of attacks targeting their assets, enabling the development of customized alerting and response strategies.

The Azure Sentinel Monitoring and Alerting Platform (MAP) further augments the organization's security capabilities by providing a live visualization of cyber-attacks as they occur. The MAP dashboard offers a real-time global map that displays markers indicating the location of attacks, empowering the organization to swiftly identify and respond to potential threats. By leveraging the honeypot in conjunction with Azure Sentinel MAP, hackers are enticed to attempt logins on a virtual machine with disabled firewalls, allowing for comprehensive logging of their location and IP information. This data is then processed through a third-party API to obtain geolocation details, which are then visualized on the MAP dashboard, providing a live view of attack locations and frequencies.

The utilization of Azure Sentinel MAP alongside the honeypot offers invaluable insight into the nature and frequency of cyber-attacks directed at the organization. This information enables the creation of custom alerts and responses to fortify defenses against future attacks. The real-time visualization capabilities of Azure Sentinel MAP empower security teams to swiftly identify and respond to threats, mitigating the potential damage caused by cyber-attacks.

Ultimately, the primary objective of this project is to bolster the organization's ability to safeguard their assets and data against cyber threats, minimizing the risks of financial loss, reputational damage, and legal liabilities. Azure Sentinel, as a cloud native SIEM solution, provides a comprehensive and versatile approach to security operations. By seamlessly collecting, analyzing, and acting upon security data from various sources, Azure Sentinel equips organizations with the necessary tools and insights to combat the evolving landscape of cybersecurity threats. Additionally, the integration with Microsoft Threat Intelligence ensures access to real-time information about the latest threats and vulnerabilities, further enhancing the organization's security defenses.

REFERENCES

- [1] Exabeam. "Machine Learning for Cybersecurity," Exabeam, <https://www.exabeam.com/information-security/machine-learning-for-cybersecurity/>.
- [2] IBM Security. "What is Security Intelligence?" YouTube, 9 Aug. 2016, <https://www.youtube.com/watch?v=YAMZ5IG5GK0>.
- [3] Levin, Yechiel, et al. "What Is Microsoft Sentinel?" Microsoft Learn, Microsoft, 13 Mar. 2023, <https://learn.microsoft.com/en-us/azure/sentinel/overview>.
- [4] Norton Online, "What is a honeypot? how it is used in cyber security?," Norton, <https://us.norton.com/blog/iot/what-is-a-honeypot>.
- [5] "Siem tutorial for Beginners | Azure sentinel tutorial map with live cyber attacks!," YouTube, <https://www.youtube.com/watch?v=RoZeVbbZ0o0>
- [6] What is Security Information and Event Management (SIEM)? (no date) IBM. Available at: <https://www.ibm.com/topics/siem>.
- [7] Yelevin, "What is Microsoft Sentinel?," Microsoft Learn, <https://learn.microsoft.com/en-us/azure/sentinel/overview> (accessed May 11, 2023).